

Ankit Chahal

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Professional Summary and Technical Expertise

- AI engineer with 5 years of experience designing production-grade ML systems, architecting LLM-powered applications, and leading MLOps and model reliability at scale.
- Skilled in prompt engineering, prompt governance, responsible AI, secure LLM applications, and AI system evaluation and guardrails.
- Strong understanding of cloud ML and data ecosystems, including AWS (S3, EC2, RDS), Azure ML, Spark-based data processing, and distributed workflows.
- Worked across industries including BFSI, FMCG, and healthcare, with hands-on experience in applied AI, model monitoring, and production pipelines.

Work Experience

Data Scientist 2

Delhi, India

Trident Information Systems

August 2023 – Present

- Led the upgrade of the VIS platform from v1.0 to v2.0, architecting multi-camera parallel streaming, redesigning the annotation engine, and integrating PLCs and industrial hardware for real-time control.
- Architected a cloud-native MLOps platform on **Azure**, leveraging **Kubeflow Pipelines** for YOLOv8 training orchestration and **MLflow** for experiment tracking and model governance.
- Built and automated CI/CD pipelines using **GitHub Actions** to deploy containerized ML microservices to **Azure Container Apps**.
- Implemented a RAG-based intelligence layer over historical breakdown data using vector search, reducing issue resolution cycles by 30% and improving operator decision latency.
- Deployed edge-based computer vision anomaly detection on live manufacturing lines, sustaining 100 parts/min throughput with hardware-level fail-safe integration.
- Streamlined ML data versioning and reproducibility using **DVC** across **AWS S3** and **Azure** storage.

Data Scientist

Delhi, India

AgriWatch

April 2023 – August 2023

- Automated financial market reporting by designing an LLM API pipeline to generate market summaries, reducing manual effort by 70%.
- Engineered a GeoDjango-based scraping and processing pipeline to build a 50-year historical dataset for crop yield forecasting, improving integration efficiency by 40%.
- Implemented governance and monitoring dashboards to track data drift and model performance for satellite imagery classification systems.

Associate Data Scientist

Bangalore, India

Lincode PVT Limited

March 2022 – March 2023

- Owned deployment of a real-time defect detection system using YOLOv5, achieving 0.95 mAP with 40ms inference latency on industrial inspection lines.
- Containerized FastAPI microservices using Docker and deployed on AWS EC2 with NGINX load balancing for the LIVIS inspection platform.
- Standardized the ML lifecycle using MLflow for experiment tracking and model versioning across deployments.

Education

Master of Science in Data Science

Chandigarh, India

Chandigarh University

July 2020 – June 2022

Bachelor of Technology in Computer Science

Mullana, Haryana, India

Maharishi Markandeshwar University

July 2015 – June 2019

Skills

Generative AI: LLMs, RAG, Azure OpenAI, LangChain, Hugging Face, LLaMA, Mistral, Claude, Chroma DB.

Machine Learning: Data preprocessing, feature engineering, model development, statistical modeling, PCA.

Deep Learning: TensorFlow, Keras, CNNs, RNNs, LSTMs, GRUs, autoencoders, GANs, model optimization.

Computer Vision: OpenCV, object detection (YOLO, SSD, Faster R-CNN), segmentation (U-Net, Mask R-Net), anomaly detection. Programming: Python, FastAPI, SQL, messaging Cloud and MLOps: AWS (S3, EC2, RDS, IAM), Azure ML, distributed data processing, model monitoring, CI/CD for ML.